Electronics - Thailand

Thailand rides the 800VDC AI wave

Industry Overview

AI infrastructure enters its next upgrade cycle

The AI buildout is moving beyond chips. As systems become larger and rack power rises from tens of kilowatts toward 600kW and eventually 1MW-class systems, the bottlenecks are shifting to the infrastructure around the compute. Power delivery, cooling and data movement now need to scale at the same pace as the chips themselves.

800VDC and optical connectivity become the next enablers

We see two major technology shifts. 800VDC addresses the power problem by reducing current, copper use and conversion losses as rack power rises. Optical connectivity addresses the data problem as copper becomes harder to scale at higher speeds and longer distances. Together, these changes expand the opportunity beyond GPUs into analog ICs, power semiconductors, power racks, cooling, facility infrastructure and photonics.

Benefits begin before full 800VDC adoption

Data centers are expected to upgrade gradually, since replacing everything at once would be too costly and disruptive. They start with the most urgent step, improving power equipment around AI racks so these can handle more electricity, then use 800VDC more widely as a more efficient way to deliver power and redesign the facility later. Thailand can benefit throughout this transition, as each stage requires more power electronics, cooling systems, semiconductor assembly, complex PCBs and other supporting hardware already linked to the country's manufacturing base.

Thailand is better-positioned than it first appears

Thailand is not a leading-edge chip-design or wafer-fabrication hub, but it has a meaningful role in the downstream supply chain. The country already has manufacturing exposure across power electronics, semiconductor assembly and testing, complex PCBs, optical products, storage, transformers and cooling fluids. As more value shifts into the hardware around AI chips, this manufacturing base becomes increasingly relevant.

Stock implications extend across Thai and foreign equities

For stock implications, we look at all Thai-listed companies that could gain benefit from 800VDC AI wave. Thai names include DELTA, HANA, KCE, SMT, CCET, AMATA, WHA PSP and TRT.

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AI: Artificial Intelligence
AC: Alternating Current
DC: Direct Current
AC/DC: Alternating Current to Direct Current
PSU: Power Supply Unit
VRM: Voltage Regulator Module
PDU: Power Distribution Unit
UPS: Uninterruptible Power Supply
ESS: Energy Storage System
BESS: Battery Energy Storage System
SST: Solid-State Transformer
MV: Medium Voltage
LV: Low Voltage
HVDC: High-Voltage Direct Current
800VDC: 800-Volt Direct Current
IBC: Intermediate Bus Converter
BBU: Battery Backup Unit
PCS: Power Conversion System
CDU: Coolant Distribution Unit
DLC: Direct Liquid Cooling
GPU: Graphics Processing Unit
CPU: Central Processing Unit
DPU: Data Processing Unit
NIC: Network Interface Card
HBM: High Bandwidth Memory
PCB: Printed Circuit Board
EMS: Electronics Manufacturing Services
Si: Silicon
SiC: Silicon Carbide
GaN: Gallium Nitride

Refer to important disclosures on page 35 to 38. Analyst Certification on page 33. Price Objective Basis/Risk on page 33.
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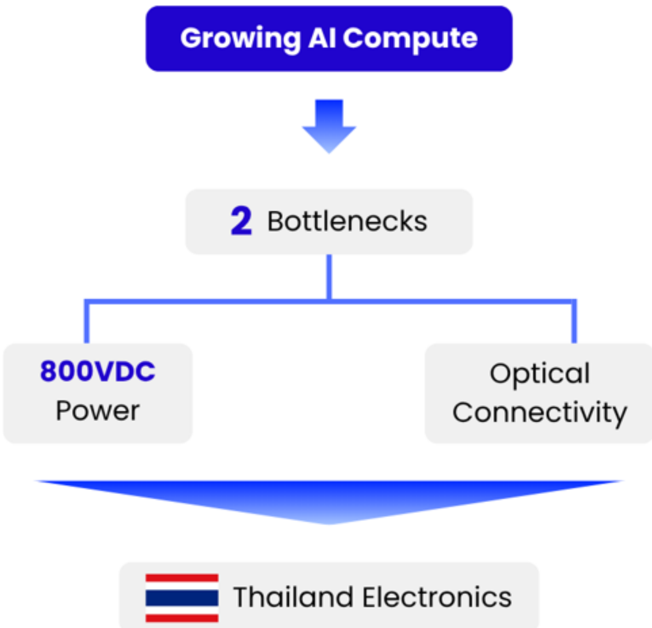
Executive summary

Exhibit 1: AI's Two Bottlenecks: 800VDC Power and Optical Connectivity

AI compute growth is pushing demand into power and connectivity, creating new opportunities for Thailand's electronics



Source: KKPS

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AI infrastructure is moving into its next bottleneck

AI investment has moved from GPUs to memory and is now reaching a new constraint: the infrastructure around the chips. As AI systems become larger, rack power is rising from tens of kilowatts toward 600kW and eventually 1MW-class systems. At the same time, more chips need to exchange much larger amounts of data. In our view, this creates two parallel upgrade cycles in data centers: 800VDC for power delivery and optical connectivity for data movement.

800VDC expands the opportunity across the whole power chain

Traditional data-center power systems were built for much lower rack power. As power rises, pushing more current through the system requires more copper and creates more heat loss. 800VDC addresses this by using higher voltage and lower current, while gradually moving power conversion away from the compute rack toward dedicated power racks and, longer term, facility-level infrastructure. The transition should therefore increase demand across server-board power, PSUs, power racks, cooling, energy storage, protection systems and new technologies such as SSTs and SSCBs. BofA estimates the AI analog semiconductor market will likely grow from US$7.9bn in CY25 to around US$28bn by CY30.

Optics address the second constraint as copper reaches its limits

Power is only one part of the problem. Larger AI systems also require much more data to move between chips and racks. Copper remains suitable for short distances but becomes harder to scale as data rates rise because of signal loss, reach and power constraints. This is driving greater use of optical transceivers, silicon photonics and other optical networking technologies. BofA estimates optical connectivity TAM could reach around US$88bn by 2030, or roughly 80% of AI connectivity spending.

Thailand could benefit through downstream manufacturing

Thailand does not need to manufacture the leading AI chip to participate in this cycle. Its opportunity sits in the hardware around the chip, including power electronics, semiconductor assembly and testing, complex PCBs, optical products, storage, transformers and cooling fluids. This gives the Thai electronics sector exposure to both the power and connectivity upgrades discussed in this report.

For stock implications, Thai-listed equities include HANA, DELTA, KCE, SMT, CCET, WHA, AMATA, PSP and TRT.

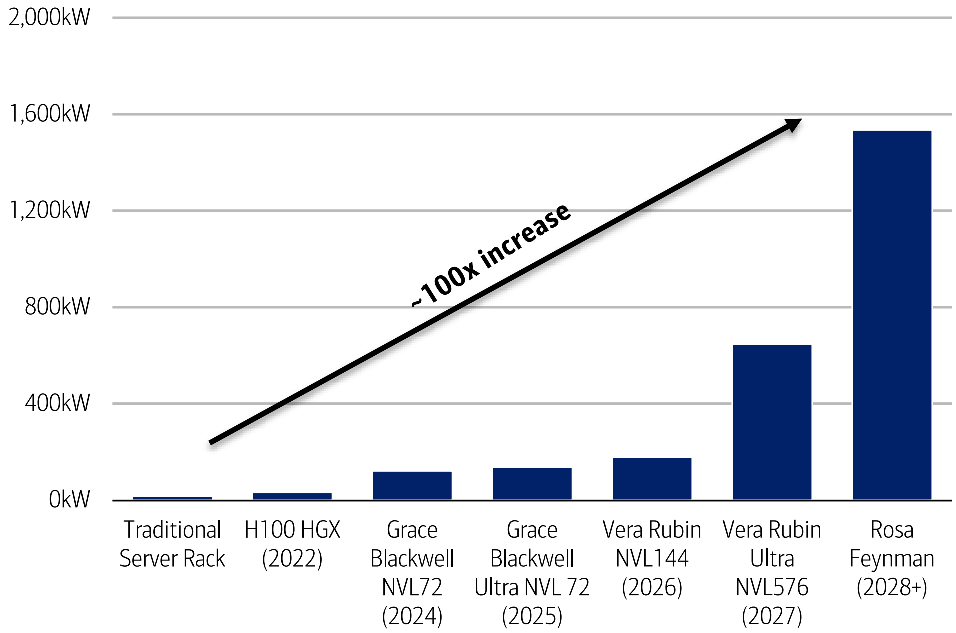
AI's insatiable appetite for power

AI investment has moved through distinct waves, each defined by a different bottleneck. The first was the GPU, the engine of AI computation. Attention then shifted to memory, as high-bandwidth memory (HBM) became the constraint on how fast those GPUs could be fed. With each bottleneck addressed, the next has come into view. Today it is the electricity these systems consume, and the heat that comes with it.

A single rack of Nvidia's newest AI servers will soon draw more electricity than a thousand American homes. A standard server rack draws 10 to 15 kilowatts, closer to what a dozen homes use. The gap is nearly a hundredfold, and it points to where the AI race is really headed. It is not a contest for chips so much as a contest for power.

Exhibit 2: Rack power across Nvidia generations

Power capacity per rack (kilowatts) across various generations of Nvidia platforms vs. a traditional server rack



Source: BofA Global Research estimates, Nvidia, Company reports

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The power demand is no longer just about the GPU. As AI evolves, it needs more types of work to be done. First, AI is trained (this means building the model). Then it is used to answer questions (this is called inference). Now, AI is also starting to handle more complex tasks, where it breaks a problem into steps and uses tools to solve it.

Because of this, the system needs more supporting parts. These include CPUs (which manage and control the system), switches (which move data between chips), network cards (which connect the system to other machines), and memory (which stores data close to where it is used).

The GPU remains the main compute engine, but it cannot work efficiently without this surrounding hardware. In the densest configurations, networking switches alone can account for close to one-fifth of total rack power. The implication is that AI power demand is becoming a full-rack issue, not just a GPU issue.

Exhibit 3: Nvidia power budget analysis
Power budget broadens across the rack

Component (in kW)	H100 (2022)	GB200 Blackwell (2024)	Vera Rubin (2026)	Rubin Ultra (2027)	Feynman (2028+)
GPU	22.4	86.4	129.6	518.4	1036.8
CPU	2.4	10.8	12.6	25.2	100.8
Switch	3	14.4	19.8	54.5	288
NIC	0.8	1.8	2.2	4.3	28.8
DPU	0.2	1.4	2.7	5.4	28.8
Overhead	2.7	6	9.7	37.8	51.4
Total Rack Power	32	121	177	646	1535

Source: BofA Global Research estimates, Nvidia

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At a basic level, the constraint comes from physics. AI chips need to exchange data at very high speed, but copper links can only carry that data over short distances. To make the system work as one large compute engine, vendors have to place more AI accelerators, CPUs, networking chips and memory closer together. Performance therefore comes from higher rack density, and higher rack density comes with more power and heat.

Nvidia's move from Hopper (old generation chip model) to Blackwell (new generation chip model) shows the trade-off. Rack power density increased by more than 3x, while system performance improved by roughly 50x. The performance gain is large, but it is achieved by packing more compute and networking hardware into the same rack.

This trend is not limited to Nvidia. AMD's latest accelerators are also moving toward 1,400-1,800W per chip, pushing rack power above 100kW. In-house ASICs from Google and Amazon may be more efficient for specific workloads, but their total rack power is also rising as systems scale. Higher power density is therefore an industry trend, not an Nvidia-specific issue.

Exhibit 4: Power consumption of other merchant GPUs and custom ASIC platforms
Rising power is industry-wide, not just Nvidia

Company	Platform	Chip	Per-Chip Power (W)	Total Rack Power (kW)
Nvidia	Vera Rubin NVL144	Rubin	1800	177
AMD	Helios	MI400X/MI455X	2000	173
AMD	MI350X rack	MI355X	1400	107
AWS	Trainium 3 UltraServer	Trainium 3	1000	100
Google	Ironwood Cube	TPU v7	1000	79
Intel	Gaudi rack	Gaudi 3	1000	51

Source: BofA Global Research estimates, AMD, Intel, AWS, Google, Nvidia

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The implication for data centers is that buying more chips is no longer enough. Each rack must receive far more power than before, and the traditional power-delivery chain is reaching its practical limits. This is where the case for 800VDC begins.

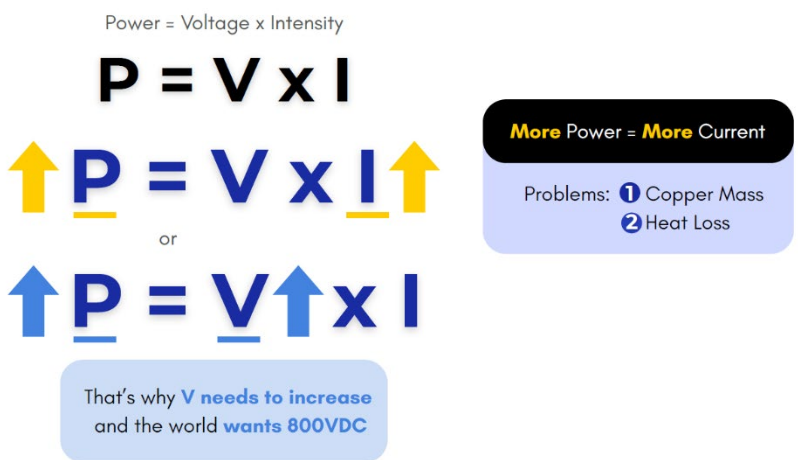
Why does the world need 800VDC?

The case for 800VDC starts with a simple equation: power equals voltage times current, or P = V × I.

Power is what the rack needs. Voltage is the pressure that pushes electricity through the system. Current (or intensity) is the amount of electricity flowing through the conductor. When rack power rises, either voltage or current must rise with it.

For many years, data centers could rely on the existing low-voltage architecture and simply push more current through the system. That worked when racks consumed tens of kilowatts. It becomes much harder when AI racks move toward hundreds of kilowatts and eventually megawatt-class power.

Exhibit 5: Power equation
Higher power consumption will end with higher voltage



Source: KKPS

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The first problem is copper. Higher current needs larger conductors, busbars and connectors. At 600kW, a legacy 48V architecture would require around 12,500 amps of current. At that level, the copper busbar can become extremely large and heavy. This is difficult to fit inside the rack and expensive to scale across a large data center.

The second problem is heat loss. Electrical losses rise with the square of current. In simple terms, doubling current can increase heat loss by around four times. This wasted energy does not create compute output. It becomes heat that the data center must remove.

800VDC solves this by raising the voltage. For the same amount of power, higher voltage means lower current. A 600kW rack at 800V requires only around 750 amps, compared with around 12,500 amps at 48V.

Exhibit 6: Benefits of 800VDC for the system
800VDC results in less copper and less heat

	Old way (54V)	New way (800VDC)	What this means?
Voltage sent through the data center	Low (54 volts)	High (800 volts)	Higher voltage moves the same power with less current
Electrical current (flow through the wire)	Very high	Much lower	Current is what creates copper bulk and heat, so less current fixes both
Copper wiring needed	Heavy, up to 200kg per rack	About 45% less	Thinner wires, less copper
Energy wasted as heat	High	Over 10x lower	Less wasted power
Bottom line	Breaks down past a few hundred kilowatts	Scales to 1MW racks and beyond	The only practical way to power today's racks

Source: BofA Global Research, Nvidia white paper

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The benefits of 800VDC can be explained in a simple sequence.
First, lower current means smaller conductors, lighter busbars and easier rack design. This reduces material usage and makes large-scale deployment more practical.

Second, electrical losses rise with the square of current. By reducing current, 800VDC significantly cuts power loss and waste heat in the power-delivery chain.

Third, this architecture is built for the next phase of AI scaling, where racks move from around 100kW today toward 600kW, 1MW and beyond, providing a clear path to expand infrastructure without hitting physical limits.

Fourth, today's power chain includes several conversion stages before electricity reaches the GPU. By reducing redundant conversion steps, 800VDC can improve end-to-end efficiency.

A simpler power chain also means fewer components and fewer failure points, which can reduce maintenance cost and improve total cost of ownership.

This is why 800VDC is not just a new electrical standard. It is the practical response to rising AI rack power. It does not make GPUs consume less energy. It makes that energy easier to deliver, with less copper, less heat loss and a power architecture that can scale with future AI racks.

The path to 800VDC: A three-stage transition

Today's AI data centers are not expected to switch directly to facility-wide 800VDC. Instead, the industry is likely to transition through three stages, with the key change being that power conversion gradually moves further upstream as rack power continues to increase.

Phase 1: Traditional AC architecture (today)

Today's data centers still follow conventional power architecture. Electricity enters the facility as AC power, passes through transformers, backup systems and power distribution equipment before reaching the compute rack. Inside the rack, AC is converted into DC and then stepped down multiple times to supply the low voltages required by GPUs.

This architecture was designed for much lower rack power. As AI racks approach hundreds of kilowatts, concentrating power conversion inside or near the compute rack consumes valuable rack space, generates additional heat and increases conversion losses. According to BofA, this architecture delivers end-to-end power efficiency of around 87.6%. See our report: Technology - Asia Pacific: HVDC for Rubin Ultra; Delta's solid R&D & position to enjoy content/shipment growth 24 November 2025.

Phase 2: Power-rack architecture (expected next step)

The first step toward 800VDC does not require rebuilding the entire data center. Instead, the main AC/DC conversion is relocated from the compute rack to a dedicated power rack positioned nearby. The existing facility infrastructure remains largely unchanged, while the power rack converts 480V AC into 800VDC before distributing power to the compute racks.

This intermediate architecture is attractive because it removes bulky conversion equipment from the compute rack, freeing up space for AI hardware while reducing heat generation close to GPUs. Since most of the upstream electrical infrastructure can remain intact, it is considerably easier to retrofit into existing facilities. BofA estimates that end-to-end efficiency improves to approximately 89.1%.

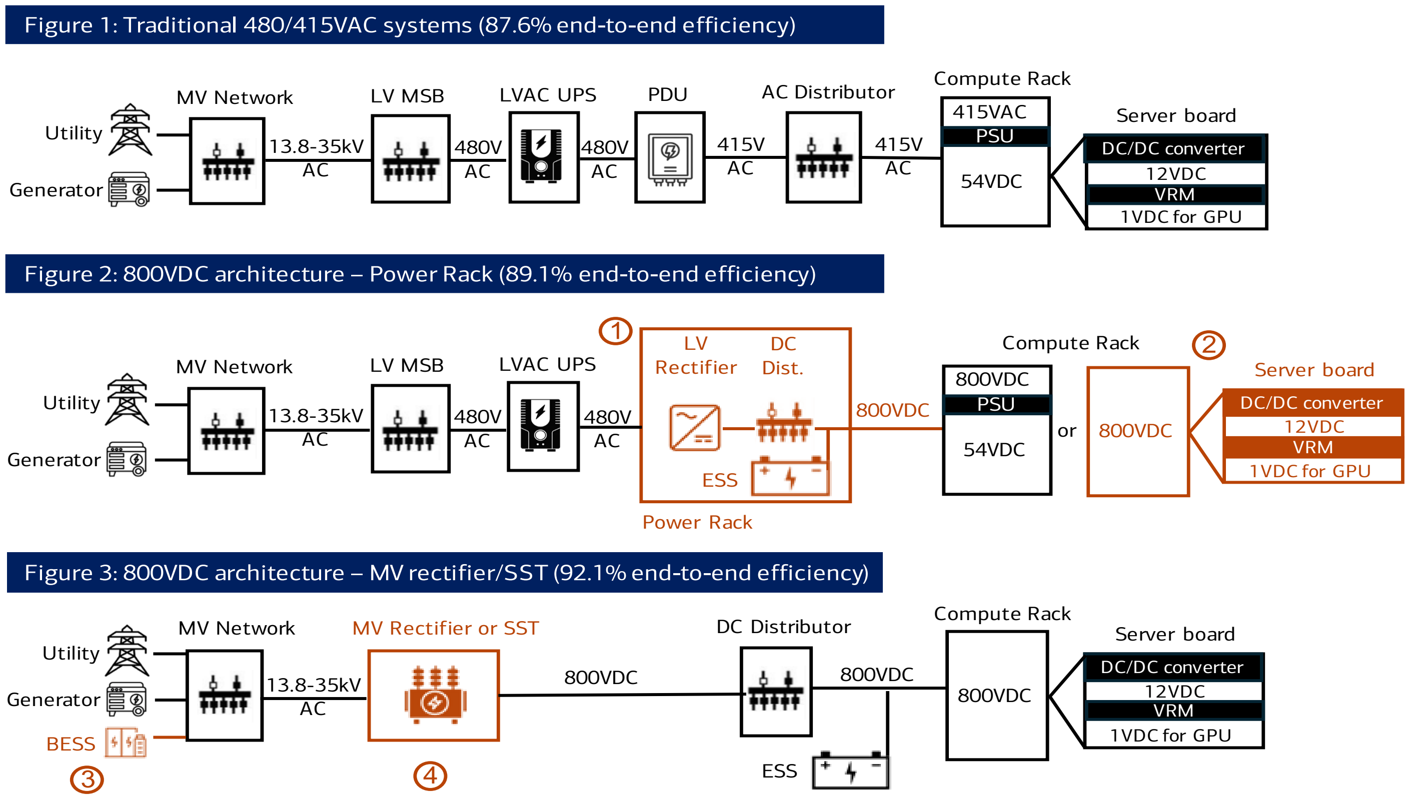
Phase 3: Full facility-level 800VDC architecture (longer term)

The final stage shifts power conversion even further upstream. Rather than distributing 480V or 415V AC throughout the facility, medium-voltage AC is converted directly into 800VDC and distributed across the entire data center with fewer conversion stages.

This architecture relies on Solid-State Transformers (SSTs), which combine wide-bandgap power semiconductors with high-frequency transformers to convert medium-voltage AC into regulated DC while improving efficiency, power quality and system control.

Although this delivers the cleanest and most efficient power architecture, it also requires the greatest redesign. Existing data centers would need new protection systems, safety standards and facility-level power equipment capable of handling high-voltage DC. SSTs themselves must also meet stringent requirements for high power, reliability and cost, making this a longer-term transition. BofA estimates end-to-end efficiency improves further to around 92.1%.

Exhibit 7: Three stages of the move from today's AC power to full 800VDC
Transition phases before the world reaches 800VDC



Key differences between the 800VDC architecture and traditional systems:
1. AC/DC conversions shifting to the power rack or the facility level
2. Less DC-DC conversions within the compute rack
3. Integration of energy storage systems to replace UPS
4. Introduction of high-frequency SSTs

Source: BofA Global Research, Nvidia white paper. AC: alternating current. BESS: battery energy storage system. DC: direct current. Dist.: Distribution. ESS: energy storage system. GPU: Graphics Processing Unit. LV: low voltage. MSB: main switchboard. MV: medium voltage. PDU: power distribution unit. PSU: power supply unit. SST: solid state transformer. UPS: uninterruptible power supply. V: voltage. VRM: voltage regulator module

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Industry roadmap

According to BofA, the industry remains firmly in Phase 1, with traditional 480/415V AC architectures still dominating today's data centers. The transition is expected to begin around 2027, when Nvidia's Rubin Ultra Kyber platform may introduce 800VDC primarily through power-rack architectures, allowing operators to improve efficiency without redesigning the entire facility.

The subsequent transition toward facility-wide 800VDC is expected to occur later. While BofA does not provide a precise timeline, its industry model introduces SST content from CY28, alongside 1MW-class AI racks and hybrid microgrid architectures, suggesting that power conversion will progressively migrate upstream as AI power density continues to increase.

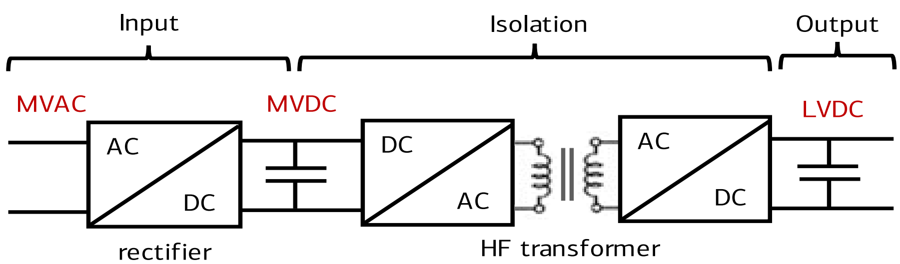
Overall, the evolution toward 800VDC is expected to be gradual rather than disruptive. The industry is likely to move first from in-rack conversion, to power-rack conversion, and ultimately to facility-level conversion, with each stage improving efficiency while requiring progressively greater infrastructure changes.

SST is the key enabler of next-generation AI-power system

SST stands for solid-state transformer. It is an electronic transformer that uses power semiconductors instead of relying mainly on copper coils and magnetic cores.

A traditional transformer is large, heavy and mostly passive. It can step voltage up or down, but it does not actively manage power flow. SST works differently. It uses advanced power semiconductors, such as SiC and GaN, together with a high-frequency transformer. This allows the transformer to be smaller, more controllable and better suited for high-voltage DC power systems.

Exhibit 8: SST structure using SiC/GaN power semiconductors and high-frequency transformer
Illustration of solid-state transformer (SST) structure



Source: BofA Global Research. AC: alternating current. DC: direct current. GaN: Gallium Nitride. LV: low voltage. MV: medium voltage. SiC: Silicon Carbide

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SST matters because the final 800VDC architecture wants to move power conversion closer to where electricity first enters the data center, at the facility level. Instead of relying on multiple older conversion steps before power reaches the AI racks, SST can help convert incoming high-voltage AC power directly into an 800VDC bus that can be distributed across the facility.

This creates three key benefits.
First, SST simplifies the power chain. By reducing the number of conversion stages, the data center can reduce power loss and remove some of the complexity of the old low-voltage architecture.

Second, SST improves control. Unlike a traditional transformer, SST is an electronic system. It can help manage voltage conversion, isolation, protection and power quality. This becomes more important as AI racks create larger and faster-changing power loads.

Third, SST supports a more DC-based data center. Once the facility has an 800VDC bus, energy storage and backup power can connect more directly to the DC system. This can reduce redundant conversion steps and make the overall power architecture cleaner.

The reason SST is not the first step is that it is technically difficult. It must handle high voltage, high power, strict safety requirements and data-center-level reliability. It also depends on advanced SiC/GaN power semiconductors, protection systems and customer qualification at scale.

The investment point is that SST pushes value further upstream in the power chain. In the old design, much of the value was concentrated in in-rack PSUs and lower-voltage conversion. In the full 800VDC architecture, more value shifts to facility-level power equipment, SSTs, protection devices, sensors, controllers and high-voltage power semiconductors. SST is therefore important because it turns 800VDC from a rack-level upgrade into a facility-level power architecture.

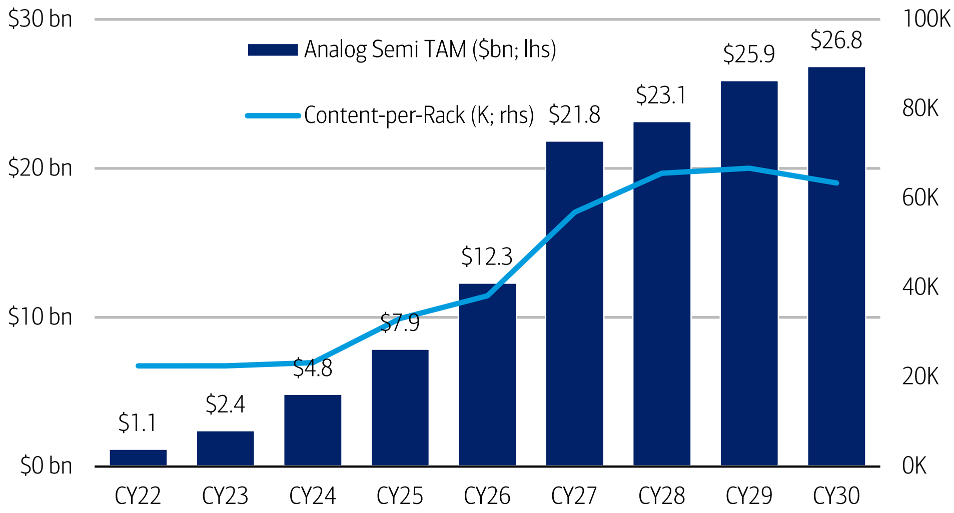
Key beneficiary sectors from the 800VDC transition

The transition to 800VDC expands the AI power opportunity well beyond server boards, creating new value pools across the entire power chain—from compute racks to facility-level infrastructure.

BofA estimates the AI analog semiconductor market will likely increase from US$7.9bn in CY25 to around US$27bn by CY30, representing a CAGR of approximately 28%. Growth is driven by both higher AI rack shipments and increasing semiconductor content per rack as power architecture becomes more sophisticated.

Exhibit 9: Analog semi TAM

The AI analog semi market should grow from US$7.9bn in CY25 to around US$27bn by CY30



Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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To better understand where this growth is captured, we examine the same US$27bn opportunity through two complementary lenses.

- **Component view** — shows where spending occurs across the AI power architecture, including server-board power, PSUs, power racks, cooling systems and facility infrastructure.
- **Semiconductor view** — shows which semiconductor technologies capture that spending, including analog ICs, silicon power devices, SiC, GaN, MCUs and sensors.

These are two perspectives on the same market. For example, a power rack appears in the component view, while the semiconductor view breaks that same power rack into its underlying chip content.

Perspective 1 — By component: Where value is created across the AI power chain

Viewed through a component lens, the key investment themes are summarized below.

Exhibit 10: Summarize table of perspective 1
Key components and why they benefit

Area/component	What it does	Why it benefits	Key number
Server-board power	Steps power down before it reaches the GPU	Higher rack power requires more board-level conversion	HV IBC content rises from US$3,000/rack at <160kW to US$216,000 at 1MW class
PSU and power racks	Convert and distribute power to the compute rack	Higher-power racks need larger and more centralized power systems	PSU analog semi TAM rises from US$1.2bn in CY25 to US$2.6bn by CY30
Cooling	Removes heat from chips, racks and facilities	More powerful racks generate much more heat	Cooling TAM rises from US$89mn in CY25 to US$477mn by CY30
Facility infrastructure	Converts, stores and protects power at facility level	800VDC adds new infrastructure such as ESS, SST and SSCB	TAM rises from US$245mn in CY25 to around US$1.8bn by CY30

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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The following sections examine each component and the key drivers behind its growth.

Component 1: Server-board power — VRMs and HV IBCs remain the foundation

Although the transition to 800VDC shifts part of the power architecture upstream, it does not eliminate the need for power conversion inside the compute rack. GPUs still operate at very low voltages, meaning power must ultimately be stepped down from the 800VDC bus to chip-level voltages through multiple conversion stages.

This is where Voltage Regulator Modules (VRMs) and High-Voltage Intermediate Bus Converters (HV IBCs) play a critical role. As rack power increases, these components must process substantially higher power while maintaining efficiency, thermal performance and power integrity. Consequently, server-board power remains one of the largest beneficiaries of AI infrastructure investment.

Exhibit 11: Evolution of HV IBC component content as racks increase in power capacity
HV IBC could reach US$216,000/rack

Component	What it means	<100kW rack	<160kW rack	600+kW rack	1MW-class rack
48V bus converter/HV IBC	Intermediate converter that steps down bus voltage before final chip-level conversion	US$1,000/rack	US$3,000/rack	US$43,000/rack	US$216,000/rack
Total rack-to-core analog content	Total analog semi content from rack input to chip-level power conversion	US$17,000/rack	US$36,000/rack	US$291,000/rack	US$917,000/rack

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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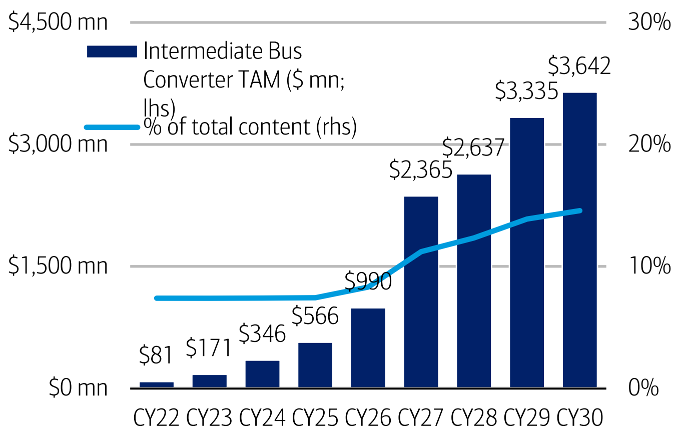
BofA's estimates show how rapidly board-level power content expands alongside rack power. Total rack-to-core analog semiconductor content increases from approximately US$36,000 per rack in today's high-performance systems to US$291,000 for 600kW racks and US$917,000 for future 1MW-class racks.

Within this, HV IBCs represent one of the fastest-growing content opportunities. Their value increases from around US$1,000 per rack in conventional systems to US$43,000 at 600kW and US$216,000 in 1MW-class architectures.

Investment takeaway: 800VDC changes where high-voltage power enters the rack, but it does not change the GPU's low-voltage power requirements. Regardless of whether power originates from a traditional AC architecture or a future 800VDC system, every GPU still requires highly efficient board-level power conversion immediately before the chip. As AI racks become more power-dense, the complexity and semiconductor content of these final conversion stages continue to increase, making server-board power one of the largest and most durable beneficiaries of AI infrastructure spending.

Exhibit 12: IBC TAM
IBC TAM could reach US$3642mn in CY30



Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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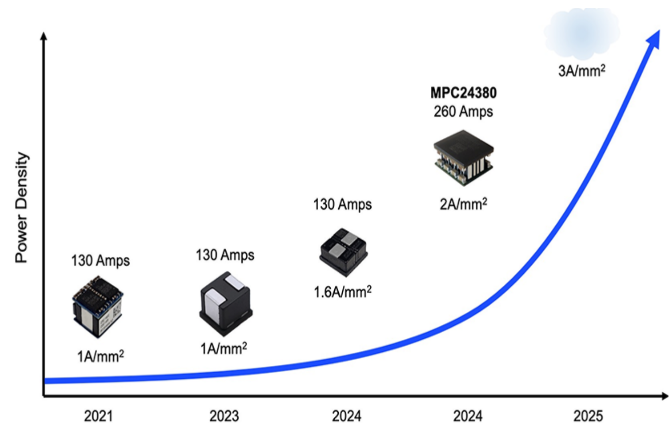
Exhibit 13: Voltage Regulator Module (VRM) roadmap
1A/mm² → 3A/mm²



Source: Monolithic Power Systems

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Component 2: PSU and power racks- value shifts from rack-level to system-level power conversion

The second major beneficiary is the power supply system. As AI rack power continues to increase, the industry is moving away from traditional rack-level power conversion toward centralized power racks capable of handling much higher power loads.

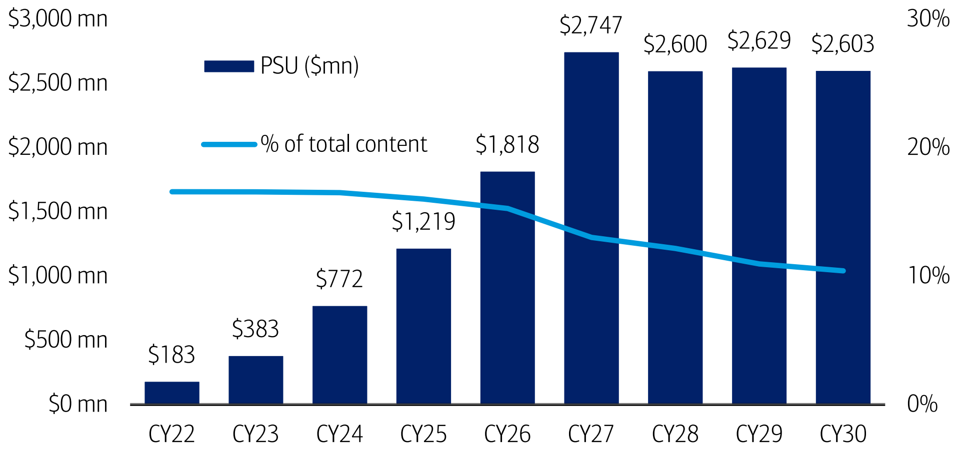
Under conventional architecture, Power Supply Units (PSUs) are located within or adjacent to the compute rack, converting incoming AC power into DC before supplying the server boards. This approach was adequate for lower-power racks but has become increasingly inefficient as rack power reaches hundreds of kilowatts, consuming valuable rack space, generating additional heat and increasing conversion losses.

The transition to 800VDC addresses these limitations by relocating front-end AC/DC conversion from the compute rack to dedicated power racks. By centralizing high-power conversion, operators can free up compute-rack space for AI hardware, improve thermal management and increase overall power efficiency.

According to BofA, the PSU-related analog semiconductor market is expected to grow from approximately US$1.2bn in CY25 to US$2.6bn by CY30. While PSUs remain an important market, the composition of spending is expected to evolve as power architecture becomes increasingly centralized.

Investment takeaway: Value is shifting from standalone rack-level PSUs toward higher-power, more integrated power racks, power shelves and system-level power management solutions. As 800VDC adoption increases, these upstream power systems should capture an expanding share of the AI power value chain.

Exhibit 14: PSU TAM
PSU total content could grow from ~US$1.2bn today to US$2.6bn by CY30



Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon,

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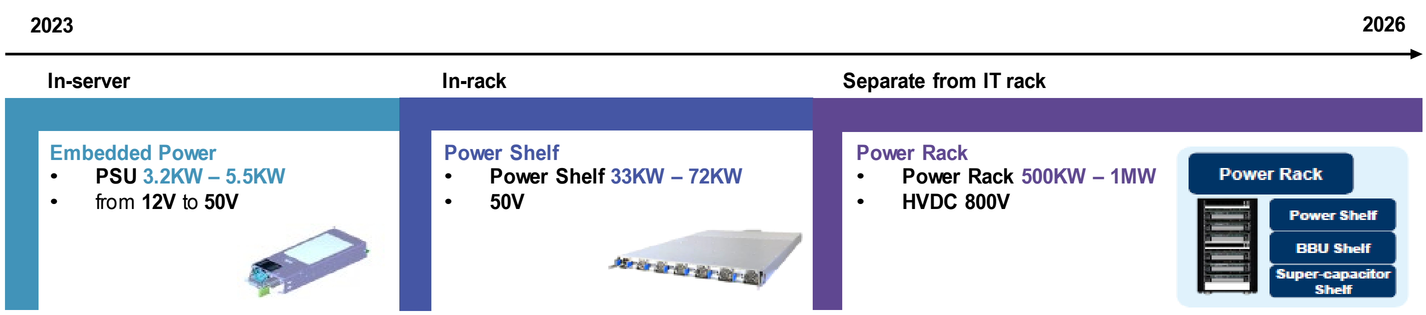
Exhibit 15: In-server PSU vs. in-rack power shelf vs. power rack sidecar
Transition phases of PSU



Source: BofA Global Research

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Component 3: Cooling and thermal management- higher power density drives greater cooling demand

Cooling and thermal management represent another key beneficiary of AI infrastructure upgrades. As rack power increases from tens of kilowatts today toward hundreds of kilowatts and eventually 1MW-class systems, removing heat becomes just as critical as delivering power.

Cooling requirements can be viewed across three layers. Chip-level cooling removes heat directly from GPUs and other processors, rack-level cooling manages heat within the server rack, while facility-level cooling rejects heat from the data hall through pumps, heat exchangers, chillers and control systems.

Traditional data centers relied primarily on air cooling because rack power was relatively modest. However, as AI racks become substantially more power dense, air cooling alone becomes increasingly inadequate. This is driving a structural shift toward liquid cooling, which can dissipate heat more efficiently and support much higher thermal loads.

800VDC and advanced cooling are complementary technologies. While 800VDC improves the efficiency of power delivery by reducing conversion losses, it does not eliminate the heat generated by increasingly powerful AI chips. As a result, advances in power delivery and cooling must occur in parallel to enable the next generation of AI infrastructure.

BofA estimates that facility cooling infrastructure content increases from approximately US$4,500/MW under traditional low-voltage architectures to around US$9,000/MW in high-voltage systems, implying that cooling content per megawatt roughly doubles as AI power density rises.

In addition, BofA projects the analog semiconductor TAM for facility cooling infrastructure to expand from approximately US$89mn in CY25 to US$477mn by CY30, reflecting growing demand for sensors, controllers, power management ICs and monitoring solutions embedded throughout modern cooling systems.

Investment takeaway: Higher rack power increases both power-delivery and thermal-management requirements. While 800VDC improves electrical efficiency, it also accelerates the adoption of more sophisticated cooling architectures, creating additional opportunities across the AI infrastructure ecosystem.

Exhibit 16: Facility cooling infrastructure value per MW table
Facility cooling content doubles in the high-voltage era

	Per MW (Low Voltage Era)	Per MW (High Voltage Era)
Components	**Sales**	**Sales**
Facility Cooling Infrastructure	US$4,500	US$9,000
Total	**US$12,350**	**US$38,920**

Source: : BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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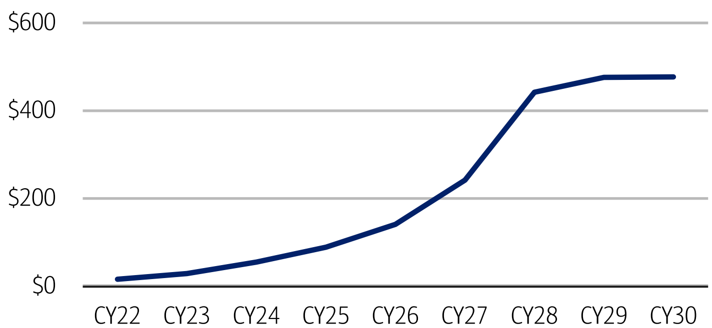
Exhibit 17: Facility cooling infrastructure TAM
Facility cooling infrastructure analog semi TAM could reach US$477mn in CY30



Source: BofA Global Research estimates, Nvidia, AMD, Intel, Marvell, Broadcom, Gartner, Omdia, company reports

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Component 4: Facility power infrastructure-a new value pool emerges (SST, SSCB, ESS and protection)

The final component pool is facility power infrastructure, which represents the most significant architectural change under a full 800VDC deployment. Unlike earlier opportunities that expand existing components, this stage creates an entirely new value pool as power conversion, storage, protection and distribution migrate upstream from the compute rack to the facility.

Key infrastructure components include:

- **Energy Storage Systems (ESS)**, which provide backup power and improve grid stability.
- **Solid-State Transformers (SSTs)**, which convert medium-voltage AC directly into the 800VDC architecture.

- **Solid-State Circuit Breakers (SSCBs)**, which provide ultra-fast fault detection and protection for high-voltage DC systems.

As discussed in the previous section, these components become increasingly important as data centers transition from rack-level power conversion toward facility-level 800VDC architectures.

Among them, SSTs are widely regarded as the key enabling technology, as they make direct medium-voltage-to-800VDC conversion possible. However, the investment opportunity extends well beyond SSTs alone. A complete 800VDC facility also requires complementary investments in ESS, SSCBs, advanced protection systems and supporting infrastructure, creating a broader ecosystem of beneficiaries.

According to BofA, the facility power infrastructure analog semiconductor market is projected to grow from approximately US$245mn in CY25 to US$1.8bn by CY30, representing a 49% CAGR. At the system level, infrastructure content is expected to increase from approximately US$12.35k/MW under traditional architecture to around US$38.92k/MW in full 800VDC deployments, reflecting the growing contribution of SSTs, ESS, SSCBs and associated infrastructure.

Investment takeaway: 800VDC expands the AI power opportunity beyond the compute rack and into facility-level infrastructure. While SSTs are the cornerstone of this transition, the long-term value creation lies across the broader ecosystem of power conversion, protection, storage and distribution technologies.

Exhibit 18: SSTs and SSCBs should gain traction from CY28 onwards
Detailed power infrastructure analog semi TAM

Detailed TAM ($; mn)	CY22	CY23	CY24	CY25	CY26	CY27	CY28	CY29	CY30
Total TAM	$44	$81	$150	$245	$385	$716	$1,706	$1,823	$1,792
Energy Storage System/UPS	$28	$51	$95	$156	$244	$407	$717	$773	$778
Solid-State Transformer	$0	$0	$0	$0	$0	$0	$304	$319	$298
Solid-State Circuit Breaker	$0	$0	$0	$0	$0	$67	$243	$255	$239
Facility Cooling Infrastructure	$16	$29	$55	$89	$141	$242	$442	$476	$477
YoY Growth (%)	CY22	CY23	CY24	CY25	CY26	CY27	CY28	CY29	CY30
Total		82.7%	85.9%	63.1%	56.9%	86.2%	138.1%	6.9%	-1.7%
Energy Storage System/UPS		82.7%	85.9%	63.1%	56.6%	66.9%	76.1%	7.9%	0.6%
Solid-State Transformer		-	-	-	-	-	-	4.9%	-6.4%
Solid-State Circuit Breaker		-	-	-	-	-	-	4.9%	-6.4%
Facility Cooling Infrastructure		82.7%	85.9%	63.1%	57.6%	72.0%	82.7%	7.7%	0.1%
Share of TAM (%)	CY22	CY23	CY24	CY25	CY26	CY27	CY28	CY29	CY30
Total	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Energy Storage System/UPS	63.6%	63.6%	63.6%	63.6%	63.4%	56.8%	42.0%	42.4%	43.4%
Solid-State Transformer	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	17.8%	17.5%	16.6%
Solid-State Circuit Breaker	0.0%	0.0%	0.0%	0.0%	0.0%	9.4%	14.3%	14.0%	13.3%
Facility Cooling Infrastructure	36.4%	36.4%	36.4%	36.4%	36.6%	33.8%	25.9%	26.1%	26.6%

Source: BofA Global Research estimates, Nvidia, AMD, Intel, Marvell, Broadcom, Gartner, Omdia, company reports

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Exhibit 19: 1MW racks will introduce new infrastructure-level technologies including SSTs and SSCBs. Content per MW could rise over 3x to US$38,900 by the end of the decade
Strategic power infra-analog semi content evolution from low voltage to high voltage

	Per MW (Low Voltage Era)		Per MW (High Voltage Era)	
Components	Sales	%	Sales	%
Energy Storage System/UPS	$7,850	64%	$14,130	36%
Solid-State Transformer	$0	0%	$8,772	23%
Solid-State Circuit Breaker	$0	0%	$7,018	18%
Facility Cooling Infrastructure	$4,500	36%	$9,000	23%
Total	$12,350	100%	$38,920	100%

Source: BofA Global Research estimates, company reports, Gartner, Omdia, Infineon, Nvidia

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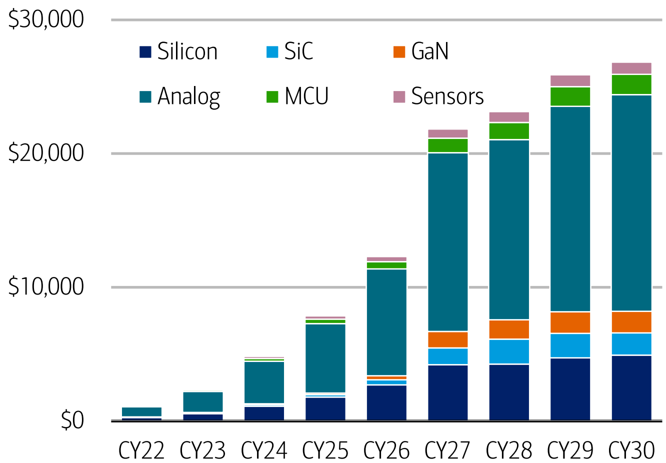
Perspective 2 — By device type: Which semiconductor technologies capture the value?

The second perspective examines the AI power opportunity by semiconductor device type rather than by system component. Whereas the previous section showed where value is created across the power chain, this section identifies which semiconductor technologies capture that value.

Every power component—whether a PSU, VRM, HV IBC or power rack—is built from multiple semiconductor devices. Together, these devices perform three essential functions: controlling power, switching power and monitoring the system.

Exhibit 20: AI analog semi TAM by device type

AI analog semi TAM should continue to see exponential growth



Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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Exhibit 20 highlights three key themes. Analog ICs remain the largest device category, reflecting their ubiquitous role throughout the power chain. Power semiconductors, particularly SiC and GaN, deliver the fastest growth as higher-voltage architecture becomes more prevalent. Meanwhile, MCUs and sensors continue to expand as increasingly complex power systems require greater coordination, monitoring and protection.

These observations are summarized below.

Exhibit 21: Summarized table of perspective 2

Key device types and why they benefit

Area/device type	What it does	Why it benefits	Key number
Analog ICs	Control, monitor and protect power	More power stages require more control and protection	TAM rises from US$5.2bn in CY25 to US$15.9bn by CY30
Silicon power devices	Switch and convert electricity	Remain widely used across mature power stages	TAM rises from US$1.8bn in CY25 to US$4.8bn by CY30
SiC	Handle high-voltage power switching	Benefits from 800VDC, power racks and SST	63% CAGR in CY25-30
GaN	Enable fast and compact power conversion	Benefits from dense and high-frequency DC/DC conversion	69% CAGR in CY25-30
MCUs	Coordinate and control power modules	More complex systems need more coordination	36% CAGR in CY25-30
Sensors	Measure voltage, current and temperature	More monitoring is needed for safety, faults and cooling	29% CAGR in CY25-30

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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Device type 1: Analog ICs - The control layer remains the largest opportunity

Analog ICs form the control layer of the AI power system. Rather than carrying the main power load, they regulate, monitor and protect power flow by measuring voltage, current and temperature while coordinating power conversion and fault protection.

Because virtually every stage of the power chain requires control and sensing, analog ICs remain the largest semiconductor category despite the emergence of new power architectures. PSUs, VRMs, HV IBCs, power racks, protection systems and cooling infrastructure all depend on analog devices to operate safely and efficiently.

Exhibit 22: Analog ICs TAM

The TAM could grow by 25% CAGR (CY25-CY30)

Semiconductor type	Role in today's data center	Role in 1MW/800VDC racks	Key vendors/companies	AI TAM CY25 (US$ mn)	AI TAM CY30 (US$ mn)	% of TAM CY25	% of TAM CY30	CY25-30 CAGR
Analog ICs	Control, sense and protect power flow	More control, monitoring and protection across the whole power chain	TXN, ADI, Infineon, MPWR, Renesas	5,198	15,919	66%	59%	25%

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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According to BofA, the AI analog IC market is expected to expand from US$5.2bn in CY25 to US$15.9bn by CY30, representing a 25% CAGR. Although faster-growing technologies such as SiC and GaN gradually increase their market share, analog ICs remain the largest value pool because every power stage requires increasingly sophisticated control, monitoring and protection.

Investment takeaway: Higher-voltage AI power systems require not only stronger power devices but also substantially more intelligence, making analog ICs the foundation of the AI power chain.

Device type 2: Power semiconductors - The fastest-growing beneficiary

Power semiconductors are devices that physically convert and switch electricity throughout the AI power chain. Unlike analog ICs, which control power flow, power semiconductors directly handle the electrical load.

Traditional silicon devices continue to dominate mature, lower-voltage applications because they are cost-effective and well-established. However, as AI infrastructure migrates toward higher voltages and greater power density, wide-bandgap semiconductors become increasingly attractive.

Silicon Carbide (SiC) is well suited to high-voltage, high-reliability applications such as power racks, SSTs and 800V protection systems, while Gallium Nitride (GaN) excels in high-frequency, compact DC/DC conversion closer to the compute tray.

Exhibit 23: Power semiconductor TAMs
The TAM could grow by 69% CAGR (CY25-CY30)

Role in today's data center	Role in 1MW/800VDC racks	Key vendors/ companies	AI TAM CY25 (US$ mn)	AI TAM CY30 (US$ mn)	% of TAM CY 25	% of TAM CY 30	CY25-30 CAGR
Basic power switching in mature power stages	Still used in lower-voltage and cost-sensitive conversion	Infineon, ON, STMicro, Renesas, TXN	1,791	4780	23%	18%	22%
High-voltage power switching	Benefits from 800V protection, power racks and SST	Infineon, ON, STMicro, Wolfspeed	183	2,098	2%	8%	63%
Fast and compact power switching	Benefits from dense DC/DC conversion near compute	Infineon, Navitas, Power Integrations, EPC	118	1,612	2%	6%	69%

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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BofA projects SiC and GaN to be the fastest-growing semiconductor categories within the AI power chain, with CAGRs of approximately 63% and 69%, respectively. Although silicon remains the largest power semiconductor market, the mix is expected to shift progressively toward wide-bandgap technologies as 800VDC architectures gain adoption.

Investment takeaway: Silicon remains the foundation today, but SiC and GaN are expected to capture an increasing share of future AI power investment.

Device type 3: MCUs and sensors - Enabling intelligent power management

MCUs and sensors form the supporting layer of the AI power system. MCUs coordinate communication and control across multiple power modules, while sensors continuously monitor voltage, current and temperature to maintain system reliability.

As AI racks become increasingly power-dense, real-time monitoring and intelligent control become more critical. Higher-voltage architecture introduces more distributed power stages, requiring additional sensing, diagnostics and system coordination to ensure safe and reliable operation.

Exhibit 24: MCU and sensors TAM
TAM CAGR (CY25-CY30) could grow by 36% for MCU and 29% for sensors

Semiconductor type	Role in today's data center	Role in 1MW/800VDC racks	Key vendors/ companies	AI TAM CY25 (US$ mn)	AI TAM CY30 (US$ mn)	% of TAM CY25	% of TAM CY30	CY25-30 CAGR
MCU	Small chips that coordinate power modules	More coordination as the power system becomes distributed	Renesas, Microchip, Infineon, STMicro, NXP	332	1,516	4%	6%	36%
Sensors	Measure voltage, current and temperature	More real-time visibility for safety, faults and cooling	ADI, TXN, Infineon, STMicro, ON	243	867	3%	3%	29%

Source: BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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Although MCUs and sensors represent smaller markets than analog ICs and power semiconductors, BofA expects them to grow at healthy CAGRs of 36% and 29%, respectively, reflecting the increasing complexity of AI power systems.

Investment takeaway: Higher-power AI infrastructure requires not only better power conversion but also smarter system management, making MCUs and sensors increasingly important enablers of next-generation data centers.

Optical connectivity is another major tailwind from AI infrastructure growth

The rapid advancement of AI models is driving a sharp increase in data exchanged between GPUs. Modern AI training requires hundreds or thousands of GPUs to operate as a unified system through high-speed scale-up interconnects.

Today, these networks primarily rely on electrical (copper) interconnects. However, as transmission speeds increase to 400G, 800G, and eventually 1.6T, copper experiences greater signal attenuation, crosstalk, and power loss, limiting high-speed transmission to very short distances.

As NVIDIA scales from old technology to much larger systems, more GPUs need to communicate with each other at very high speeds. Copper works well over short distances, but its performance drops as distance and data rates increase. This makes it harder to connect larger GPU clusters using copper alone. Optical interconnects help solve this problem by carrying more data over longer distances with lower signal loss, supporting the move toward larger AI systems.

Optics allow AI systems to scale beyond the limits of copper

As AI systems grow, putting more and more GPUs close together becomes harder because of space, power and cooling limits. Copper links also work best over short distances. Optical links use light to move data, allowing large amounts of data to travel farther with less signal loss.

Google's Optical Circuit Switching, or OCS, is one example. OCS is an optical switch that connects different groups of AI chips using light. When one group needs to exchange a large amount of data with another group, the switch opens a direct optical path between them.

This makes it easier to connect many groups of AI chips into one larger system without relying only on short copper links. Google's Ironwood Superpod, for example, can scale to 9,216 TPUs. The key point is that optics are becoming more important as AI clusters grow because they allow more chips to communicate across a larger system.

Exhibit 25: Optical networking enables significantly larger AI cluster

Google's optical networking architecture supports up to 9,216 TPU chips per pod, compared with 576 GPUs in NVIDIA's current scale-up domain

Metric	Nvidia B300/GB300 "Blackwell Ultra"	Google TPU v7 "Ironwood"
HBM capacity	288 GB HBM3e per GPU (12 stacks)	192 GB HBM3e per chip (8 stacks)
HBM bandwidth	8 TB/s per GPU	7.4 TB/s per TPU
# chip per rack	72x	64x
# rack per pod	8x	144x
# chip per pod	576x	9,216x

Source: Nvidia, Google

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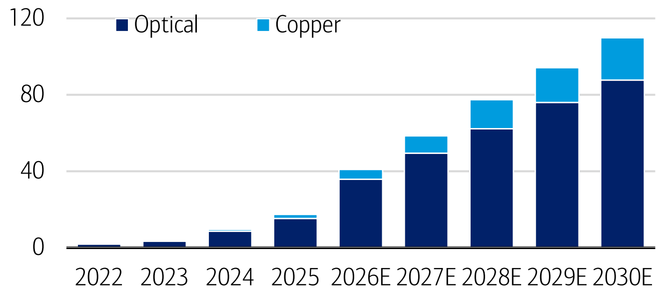
Optics are expected to dominate future AI connectivity spending

Although copper remains widely used for short-reach connections such as GPU-to-GPU and server-to-server communication, optical technologies are expected to capture the majority of future AI connectivity spending.

Bofa estimates that optical connectivity TAM will likely reach approximately US$88bn by 2030, representing around 80% of the AI connectivity market, while copper connectivity is expected to reach only US$22bn, remaining focused on short-reach applications and power delivery where optics cannot replace electrical connections.

Exhibit 26: Optical and copper TAM
Optical is in the early stage of growth



Source: BofA Global Research estimates, Gartner, Mercury Research, IDC, LightCounting, 650
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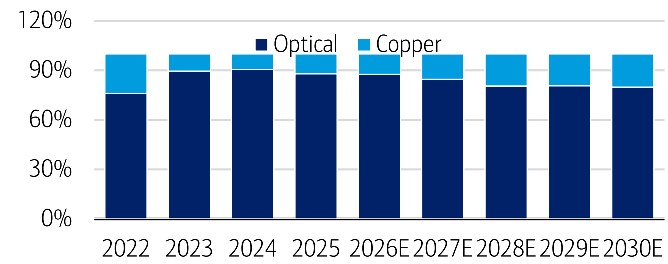
Exhibit 27: Optical market share
Optical connectivity is expected to capture the majority of AI networking TAM



Source: BofA Global Research estimates, Gartner, Mercury Research, IDC, LightCounting, 650
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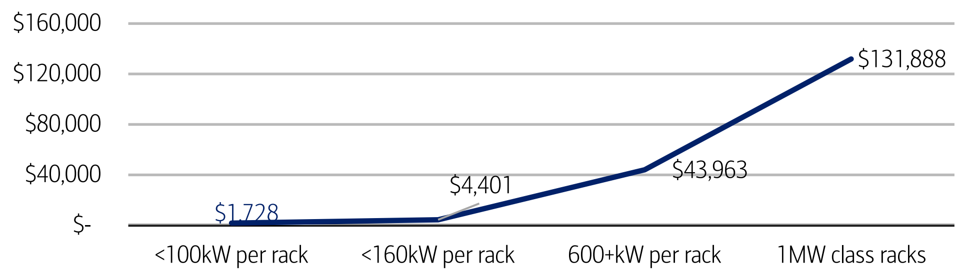
As networking speeds migrate from 800G to 1.6T and AI racks move toward the 1 MW-class envisioned in the NVIDIA's Feynman GPU generation, the amount of optical equipment required per rack is also expected to increase significantly, driving substantial growth in optical infrastructure value.

Exhibit 28: Optical infrastructure TAM
The value of optical infrastructure within a single rack is projected to surge as power capacity increases



Source: BofA Global Research
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Implications for Thai equities

Thailand's role: the factory floor of the power revolution

Thailand is building a stronger role in the AI semiconductor value chain, particularly in downstream manufacturing where advanced technologies are turned into deployable hardware. While leading-edge fabs, chip design, SiC/GaN technology and photonics IP remain concentrated in Taiwan, Korea and the US, Thailand's growing capabilities position it well to capture increasing demand in this segment.

That fits well with the themes discussed earlier. Higher-power AI racks need more power electronics, denser PCBs, cooling hardware, assembly and optical components. Thailand already has meaningful manufacturing exposure across these areas, so the local benefit is more about rising content around the AI chip than the chip itself.

For Thai equities, the stocks mentioned in this report include Delta Electronics (Thailand) (DELTA), Hana Microelectronics (HANA), KCE Electronics (KCE), Stars Microelectronics (SMT), Cal-Comp Electronics (Thailand) (CCET), WHA Corporation (WHA), Amata Corporation (AMATA), Tirathai (TRT), and PSP Specialties (PSP).

Implication to HANA: Downstream read-through from analog ICs and power semiconductors

HANA could benefit mainly through two parts of the AI power chain discussed earlier: analog ICs and power semiconductors. On analog ICs, HANA provides assembly, packaging and testing services for semiconductor customers, so stronger demand for power-management and control chips could support higher outsourced manufacturing activity. On power semiconductors, HANA has existing capabilities in power modules through its PMS business and is working toward SiC power-module applications for data centers. This gives the company potential exposure to the faster-growing power semiconductor segment as 800VDC and higher-power AI infrastructure develop.

To quantify the potential upside from AI tailwinds spreading across multiple end-markets, we apply BofA's 2025-30 sector growth assumptions to HANA's 2025 revenue mix.

Exhibit 29: HANA revenue upside scenario based on BofA end-market growth

Bull scenario on top of our base case

End-application	2025 year-end Product Mix %	BofA proxy end-market	Growth (CAGR '25-30E)	2025	2030E
Telecom	14%	Smartphone	0.0%	2,879	2,879
Auto	23%	Automotive	8.3%	4,729	7,046
Industrial	24%	Industrial & Other	10.6%	4,935	8,167
RFID	14%	No BofA proxy	0.0%	2,879	2,879
Opto	5%	Optical Equipment	16.9%	1,028	2,245
Consumer	9%	Consumer	1.3%	1,851	1,974
Computer	6%	Compute and Storage	19.1%	1,234	2,957
Medical	5%	No BofA proxy	0.0%	1,028	1,028
Total revenue (assume BofA proxy end-market)				**20,563**	**29,175**
Contribution to net profit (Using 2021 upcycle NPM)					2,801
Current 2030 net profit assumption					1,639
Incremental net profit					1,161
Incremental EPS					1.31
Applying targeted PER of 13x (in 2030 based on our current PO)				**Bt**	**17**
Potential upside to our 2027 PO (WACC =7.1%)				**Bt**	**14**

Source: KKPS Estimates, BofA Global Research, Mercury Research, Gartner, Omdia, SIA

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We start with HANA's 2025 revenue of Bt20.6bn and split it by end-market. For each segment, we apply the closest BofA industry CAGR through 2030. For example, we use 8.3% CAGR for automotive, 10.6% for industrial, 16.9% for optical equipment and 19.1% for compute and storage. For RFID and medical, where we do not have a direct BofA proxy, we assume no growth.

Under these assumptions, HANA's revenue could reach Bt29.2bn by 2030. We then apply HANA's 2021 net margin, using the most recent semiconductor upcycle as a reference, which gives 2030 net profit of around Bt2.8bn. This compares with our current 2030 net profit forecast of Bt1.64bn, implying incremental net profit of around Bt1.16bn, or Bt1.31 per share in additional EPS.

To translate this into valuation upside, we apply a 13x PER, which is the multiple implied by our current Bt42 PO against our 2030E EPS. Applying this multiple to the incremental 2030 EPS gives around Bt17/share of additional value in 2030, or around Bt14/share when discounted back to 2027 at a 7.1% WACC.

Risks: Our scenario depends on BofA's end-market growth assumptions. If growth in industrial, automotive, optical or compute markets is weaker than expected, the upside would be lower.

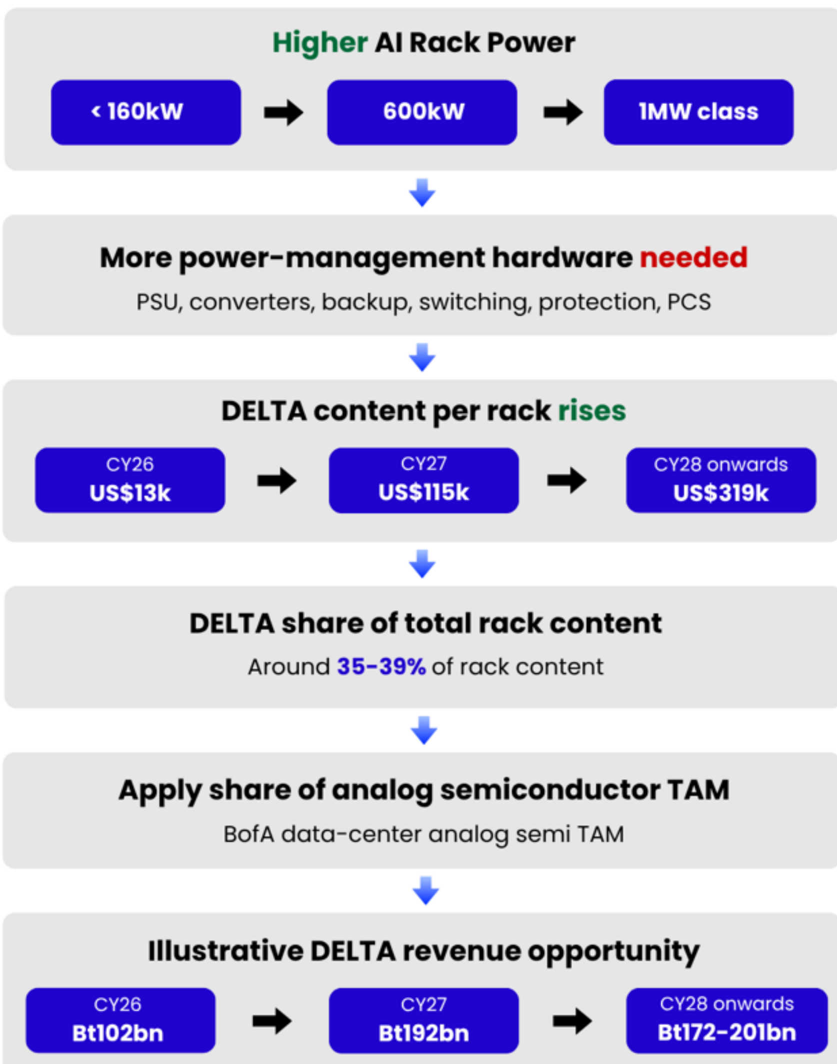
Zooming in on HANA, its existing businesses still face risks. A sharper-than-expected rise in memory prices could weaken smartphone and other electronics demand, while delays in customer qualifications or the planned SiC power-module ramp could push out new growth opportunities. Given the longer-term nature of these projects, any delay could also weigh on stock sentiment.

Furthermore, on valuation, the market is currently giving HANA a premium for the semiconductor upcycle and, increasingly, for its AI-related potential. If execution is delayed or AI-related opportunities underdeliver, the stock could face a sharp near-term correction.

Implications for DELTA (Thailand): 800VDC can materially raise content per rack

Exhibit 30: Summarization of DELTA as beneficiary

How DELTA benefits from 800VDC



Source: KKPS

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DELTA is one of the clearest Thai beneficiaries of higher-power AI racks given its exposure to both data-center power products and liquid cooling. As rack power rises, demand should increase for larger PSUs, voltage conversion, backup power, switching, protection and power management. At the same time, higher heat density should drive greater use of liquid cooling, giving DELTA another source of potential upside.

Exhibit 31: Estimated benefit of 800VDC to DELTA Thailand
DELTA could benefit from the 800VDC infrastructure

	CY26	CY27	CY28	CY29	CY30
Representative rack	<160kW	600kW	1MW	1MW	1MW
Possible DELTA content/rack	US$13,000	US$115,000	US$319,000	US$319,000	US$319,000
Share of total rack content	37%	39%	35%	35%	35%
Illustrative DELTA revenue opportunity	Bt102bn	Bt192bn	Bt172bn	Bt193bn	Bt201bn
% of 2025 DELTA Thailand revenue	52%	97%	87%	97%	101%

Source: KKPS estimates, BofA Global Research estimates, Company reports, Gartner, Omdia, Infineon, Nvidia

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The increase comes from the amount of power equipment needed as racks become larger. A lower-power rack can rely on a more conventional setup, but 600kW and 1MW-class racks need much more equipment to convert voltage, provide backup power, distribute electricity and protect the system. Based on BofA's content assumptions, the value of products that could be relevant to DELTA rises from around US$13,000 per rack in CY26 to US$115,000 in CY27 and US$319,000 from CY28 onward.

We then use this product mix to estimate the wider market opportunity. In our calculation, DELTA-related products account for around 35-39% of total rack content. We apply this percentage to BofA's data-center analog semiconductor TAM, assume DELTA captures 70% of the relevant market and use Bt33/US$. This gives an illustrative revenue opportunity of around Bt102bn in CY26, Bt192bn in CY27 and Bt172-201bn in CY28-30. The estimate does not include liquid cooling, so there could be further upside potential if demand for cooling systems rises alongside rack power and heat density.

Risks: Our scenario is only a rough estimate and assumes DELTA can maintain a large share of the relevant power market, supported by high barriers to entry, while Nvidia's platform roadmap and the 800VDC transition proceed broadly on schedule. Any delay in new architecture, slower adoption, or loss of market share would reduce the upside. Visibility is still limited, which is why we have not priced this scenario into our PO. In the near term, supply-chain issues, including component shortages and higher input costs, could also weigh on earnings and stock sentiment.

Implications for KCE (Not covered): AI raises the value of PCB complexity

KCE is likely to benefit from the AI-driven upgrade cycle through a tighter and more valuable advanced PCB market. AI servers require very high-layer PCBs to support faster data transfer, higher power and more complex system design. As global PCB leaders allocate more capacity to these high-end AI server boards, supply in other advanced PCB segments should also tighten. This creates a better industry backdrop for KCE, especially as the company continues to shift toward higher-complexity products.

KCE does not need to be a direct supplier of the highest-layer AI server PCB to benefit. The AI boom could still improve the overall PCB supply-demand balance by absorbing advanced capacity globally. This will likely reduce pricing pressure and support better mix for companies that can produce more complex boards than standard consumer PCBs.

The main product angle for KCE is HDI (High-Density Interconnect). HDI is not the same as the very high-layer PCBs used in AI servers, but both belong to the broader advanced PCB market. HDI allows more electrical connections to fit into a smaller space through finer lines, tighter spacing and laser-drilled holes. This is useful in products that need compact, high-performance electronics, such as advanced automotive systems, cameras, sensors, communication devices and automation equipment.

Management stated it plans to increase advanced-technology products from less than 10% of sales before COVID to around 40–50% over the next three to five years. The company is also moving beyond its traditional automotive base into areas such as 5G/6G, servers, automation and other higher-complexity applications.

Longer term, AI could create another layer of demand beyond data centers. Autonomous systems and humanoid robots require many sensors, cameras, control units, connectivity modules and power electronics to work together in real time. These applications should increase demand for smaller, denser and more complex PCBs, where HDI capability becomes increasingly relevant.

Therefore, KCE stands as an indirect beneficiary of the AI infrastructure cycle. The first benefit comes from tighter advanced PCB supply as AI server demand absorbs global capacity. The second comes from KCE's own mix upgrade toward HDI and other higher-complexity products.

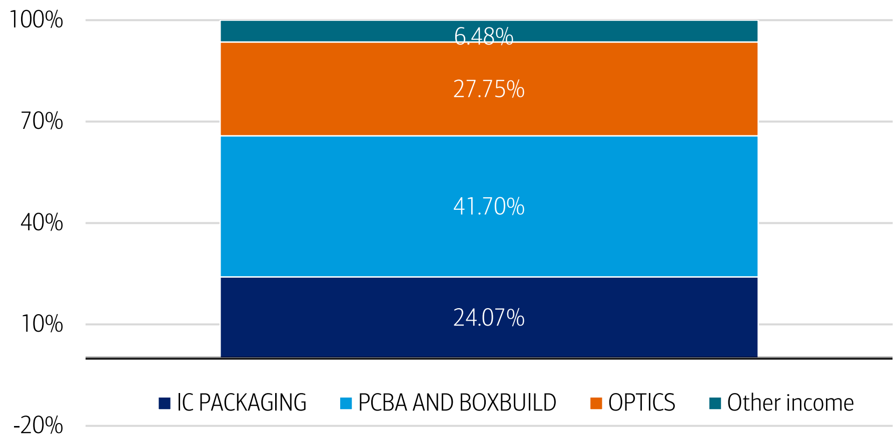
Risk: The main risk is slower-than-expected demand for advanced PCBs, particularly if AI server, 5G/6G or automation investment grows more slowly than expected. KCE also still depends heavily on automotive demand, so a weaker auto cycle could offset progress in new applications. In addition, HDI and higher-layer boards are harder to manufacture, so delays in customer qualification, slower ramp-up or lower-than-expected production yields could limit the benefit from a richer product mix

Implications for SMT (Not covered): Possible benefit from optics

SMT is well-positioned to benefit from the structural growth in AI data center optical connectivity. Nearly 30% of the company's revenue is derived from optical products, including silicon photonics components, optical transceivers, and Active Optical Cables (AOCs). As AI data centers continue to scale, demand for high-speed optical interconnect solutions is expected to increase significantly, driven by rising bandwidth requirements and the transition away from copper interconnects at higher data rates. Given its existing product portfolio, SMT is well aligned with this industry trend and stands to benefit from the growing adoption of optical connectivity.

In addition, hyperscale cloud service providers have yet to become a major customer segment for SMT. Successful penetration into this market could provide a meaningful incremental growth opportunity, supported by the increasing capital expenditure on AI infrastructure by leading hyperscalers.

Exhibit 32: Revenue breakdown
Around 27.75% of revenue came from optics in 2025



Source: SMT

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Risks: The main risk is slower-than-expected demand for advanced PCBs, particularly if AI server, 5G/6G or automation investment grows more slowly than expected. SMT also still depends heavily on automotive demand, so a weaker auto cycle could offset progress in new applications.

Implications for CCET (Not covered): Higher storage demand drives more production

For CCET, the clearest AI link is storage. As AI training and inference generate more data, data centers need more capacity across both HDDs and SSDs. This matters for CCET because its product portfolio already includes external HDDs, PCBA for hard disks, SSDs and storage-server PCBA.

CCET does not disclose storage revenue separately. According to the company's annual report, in 2025, the broader computer-peripheral segment accounted for 86.4% of total revenue, and this segment includes CCET's HDD-, SSD- and storage-related products.

The company itself has also pointed to stronger storage demand. In its 2025 annual report, CCET said demand for both traditional and solid-state storage had increased alongside cloud, data-center and AI expansion.

The implication for CCET is mainly higher manufacturing volume rather than direct exposure to memory pricing. CCET is an EMS/ODM manufacturer, not an upstream NAND or HDD technology owner. If tight supply leads storage vendors to expand output and secure more assembly capacity, CCET could benefit through higher orders for HDD PCBA, SSD and storage-related products.

Risks: The main risk is slower-than-expected adoption of optical connectivity. If the move toward higher-power architectures such as 800VDC is delayed, rack density and bandwidth requirements could rise more slowly, reducing near-term demand for optical products. A slower rollout of co-packaged optics (CPO) would also limit the expected shift from copper to optics.

Implications for Thailand's industrial sector (AMATA, WHA, NNCL, FPT and ROJNA): Upside for land sales and earnings

Thailand has launched a national semiconductor strategy to develop a complete semiconductor value chain by 2050. The strategy focuses on power semiconductors, sensors, photonics, discrete devices, and analog chips, leveraging the country's well-established electronics manufacturing base serving the automotive, energy, data center, and industrial sectors.

The government aims to attract Bt2.5tn (US$79bn) of semiconductor investment and develop 230,000 skilled workers by 2050. To support these objectives, it plans to strengthen investment incentives, expand R&D capabilities, develop semiconductor talent, upgrade infrastructure, and enhance the overall business environment.

Thailand has already gained strong investment momentum in PCB manufacturing, outsourced semiconductor assembly and testing (OSAT), hard disk drives (HDDs), and electronic components. These industries provide a solid foundation for the country to move further up the semiconductor value chain and attract higher-value manufacturing and technology investments

WHA and AMATA, Thailand's two leading industrial estate developers, are well-positioned to benefit from the expanding electronics and semiconductor supply chain. As the global AI capex cycle accelerates, we expect a new wave of FDI into Thailand, particularly in semiconductor, PCB, and advanced electronics manufacturing.

We estimate this could drive annual land sales upside of approximately 2,500 rai for WHA and 2,000 rai for AMATA. Based on these assumptions, we estimate potential earnings upside of 17-26% for WHA in 2027E-2028E and 27% for AMATA, highlighting the significant leverage of both companies to Thailand's long-term semiconductor investment cycle.

Risk: Delays in National Semiconductor Strategy execution and regional peer competition (particularly from Malaysia and Vietnam)

Exhibit 33: Thailand's industrial estate developers and semiconductor exposure
Industrial estates have a strong base from the semiconductor supply chain

	Semiconductor supply chain cluster	Key Clients
WHA	Strong AI, PCB and electronics ecosystem	Infineon, Western Digital, Foxsemicon, Compeq, Unimicron, Unitech, Google, AWS
AMATA	Large semiconductor cluster in Chonburi, supply chains and equipment	Foxsemicon (UniEQ), FT1 (HANA + PTT JV), Microchip, HANA Microelectronics, AWS
ROJNA	Leading PCB manufacturing cluster	Multi-Fineline (Dongshan Precision), Western Digital, MinebeaMitsumi
NNCL	Established semiconductor backend cluster	Lumentum, Infineon, Analog Devices, Microchip, NXP, Mitsubishi Electric
FPT	Expanding into semiconductor & high-tech manufacturing	Infineon in new JV project of Araya Industrial Estate (east of Bangkok)

Source: KKPS, Companies

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Exhibit 34: The potential upside of new land sales and earnings for WHA and AMATA
Potential FDI in Thailand from related AI CAPEX would bring significant upside to new land sales and earnings for WHA and AMATA in 2027-2028

	2027E			2028E		
New land sales (rai)	**Base case**	**Best case**	**% upside**	**Base case**	**Best case**	**% upside**
WHA	1,600	2,500	56%	1,800	2,500	39%
AMATA	600	2,000	233%	640	2,000	213%
Net profit (Bt mn)						
WHA	6,001	7,098	18%	6,036	7,637	27%
AMATA	3,230	4,100	27%	3,056	3,870	27%
EPS (Bt mn)						
WHA	0.40	0.47	17%	0.40	0.51	26%
AMATA	2.81	3.57	27%	2.66	3.37	27%
PER (x)						
WHA	12.6	10.7	-	`1.5	9.9	-
AMATA	9.2	7.2	-	9.7	7.6	-

Source: KKPS estimates

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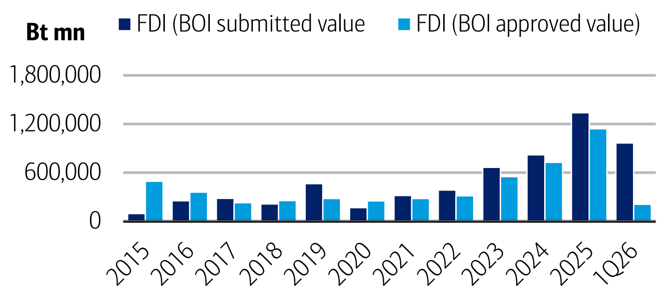
Exhibit 35: FDI in Thailand
FDI in Thailand hit an all-time high in 2025



Source: Thailand Board of Investment (BOI), KKPS
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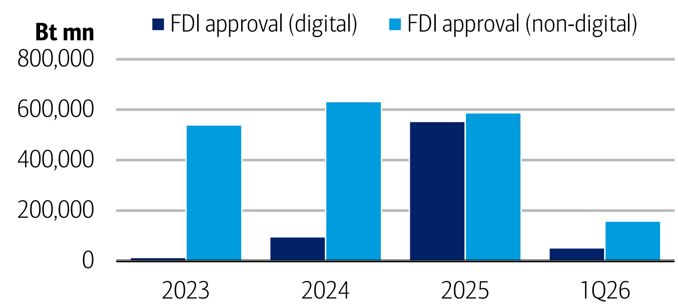
Exhibit 36: FDI in Thailand, by digital and non-digital sector
Digital sector jumped significantly in 2024-2025



Source: Thailand Board of Investment (BOI), KKPS
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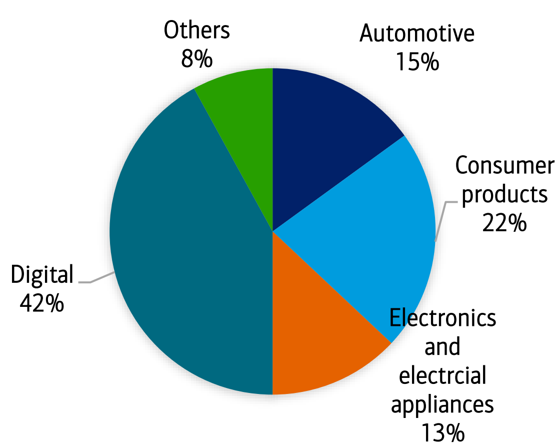
Exhibit 37: WHA – Land sales breakdown by sector for 2024-1Q26
WHA has 42% from the new economy (data centers)



Source: Company, KKPS, BofA GLOBAL RESEARCH

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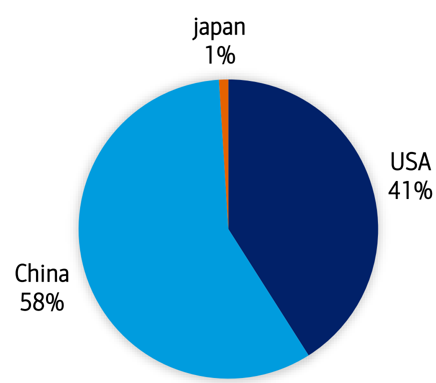
Exhibit 38: WHA – Land sales by client's nationality for 2024-1Q26
WHA's clients are balance between USA and China



Source: Company, KKPS, BofA GLOBAL RESEARCH

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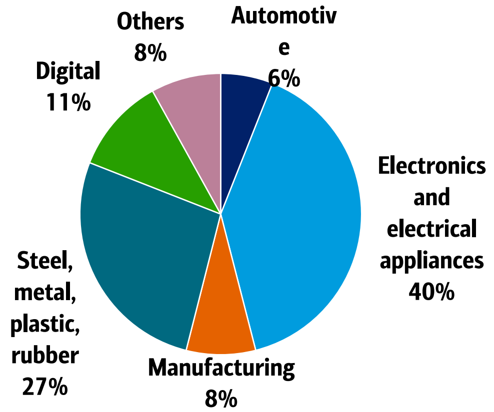
Exhibit 39: AMATA – Land sales breakdown by sector in 2025
AMATA has 11% from the new economy (data centers)



Source: Company, KKPS, BofA GLOBAL RESEARCH

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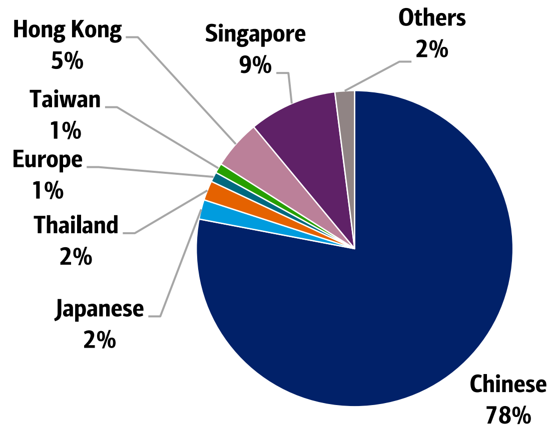
Exhibit 40: WHA – Land sales by client's nationality for 2024-1Q26
WHA's clients are balanced between the US and China



Source: Company, KKPS, BofA GLOBAL RESEARCH

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Implications for PSP (Not covered): Benefit from cooling and transformer oil

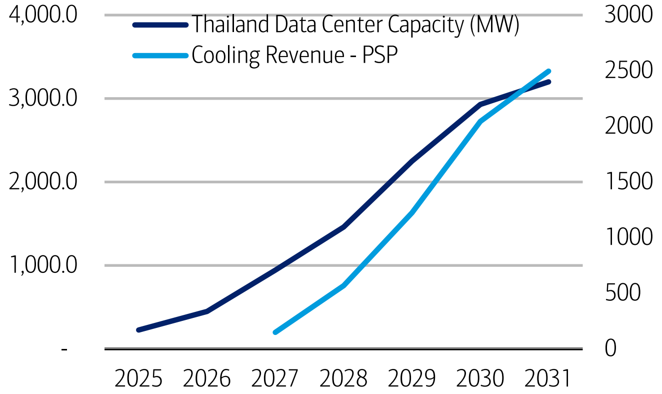
PSP could benefit from Thailand's data-center expansion through two businesses. The nearer-term opportunity is transformer oil, while liquid cooling could become a larger new growth area as rack power and heat density increase.

On transformer oil, more data-center capacity requires additional transformers and power-distribution equipment. According to management, PSP holds around 70% of Thailand's transformer-oil market and expects to capture almost all incremental domestic demand following recent supply disruption at a Shell refinery in Qatar. The company is also developing bio-based transformer oil with EGAT, MEA, PEA and GGC.

On cooling, management estimates Thailand's data-center capacity will increase from 227MW in 2025 to 2.93GW by 2030. PSP is targeting both direct-to-chip cooling, where cooling fluid removes heat directly from high-power chips, and immersion cooling, where server equipment is placed in a non-conductive fluid. Under company assumptions, DTC adoption could reach 30% by 2030 versus 9% for immersion cooling.

If PSP successfully commercializes both cooling products, company estimates suggest cooling revenue could reach around Bt2.04bn by 2030, equivalent to about 16% of FY2025 revenue, with estimated gross profit of around Bt511mn, or about 29% of FY2025 gross profit. PSP has also signed an MOU with Eco Atlas and Evonik to co-develop immersion-cooling fluids.

Exhibit 41: Thailand's data center capacity vs. estimates of potential cooling revenue for PSP
Thailand's data centers drive long-term demand for liquid cooling and expand PSP's revenue opportunity



Source: PSP

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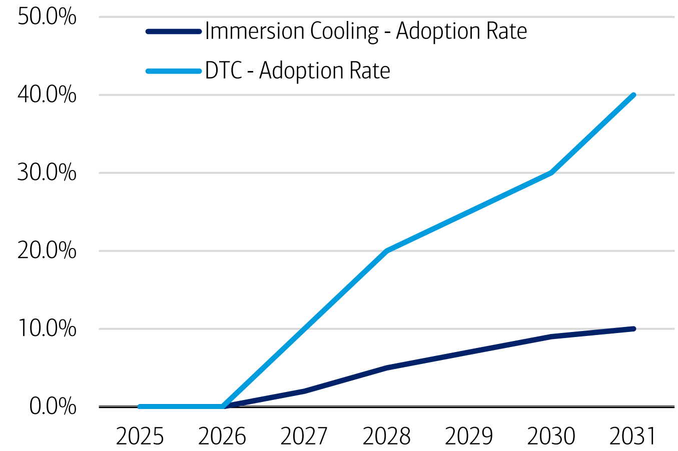
Exhibit 42: Adoption rate of immersion cooling and direct to chip cooling
The adoption rate is expected to rise



Source: PSP

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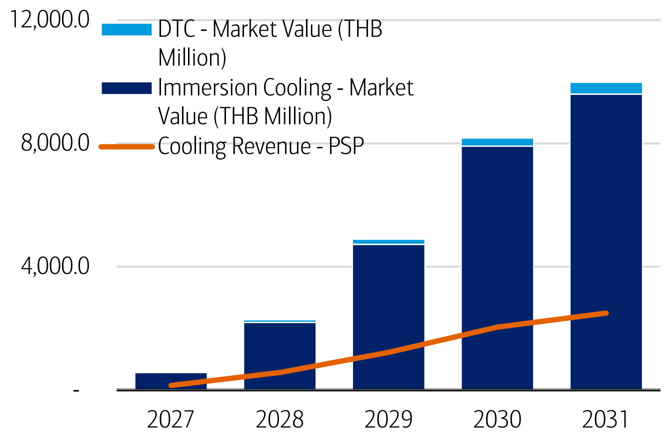
Exhibit 43: Estimated market value of PSP's cooling revenue
Both types of cooling remain the larger revenue opportunity, as PSP's revenue is still expected to represent only a fraction of the total market



Source: PSP

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Risks: The main risk is slower-than-expected adoption of liquid cooling, particularly if data-center capacity in Thailand expands more slowly than management expects or if operators continue to rely on conventional cooling for longer. Delays in commercializing PSP's immersion-cooling fluids could also push out the revenue opportunity. For transformer oil, upside could be lower if domestic demand grows more slowly or if current supply disruptions ease, reducing the incremental volume PSP expects to capture.

Implications for TRT (Not covered): Transformer shortage creates a direct AI infrastructure read-through

TRT stands to benefit from the global shortage of large power transformers. According to management, TRT is one of only around six companies in Thailand capable of producing large power transformers. Moreover, a key advantage for TRT is its role as an OEM manufacturing partner for Siemens.

The shortage first emerged in the US and Europe, where rising grid investment and data-center demand tightened transformer supply. Buyers then shifted more orders to major Asian manufacturing hubs such as Korea and Japan, but capacity there has also become constrained given lead times of around two to three years. As these markets fill up, more demand is flowing into Southeast Asia, including Thailand. TRT is already seeing this in its order book, with management reporting transformer backlog of more than Bt3.2bn and most export orders linked to its OEM partnership with Siemens. This relationship also supports a better margin profile, as TRT can participate in higher-spec export orders through Siemens' global platform, where pricing is typically stronger than for standard domestic orders.

Management expects sales to rise from Bt2.07bn last year to more than Bt3.0bn this year, followed by at least 10% growth in 2027. Utilization of large-transformer capacity is also expected to increase from around 60%-70% to full utilization before year-end.

AI data centers increase electricity demand, but that power still has to move through the grid before reaching the site. With global transformer lead times already stretched, TRT offers a direct Thailand read-through from the shortage in grid equipment, with backlog growth, higher utilization and export mix supporting both revenue and margin upside.

Exhibit 44: TRT's backlog

TRT has more than Bt3.2bn of transformer backlog on hand

in mn	Customer	Backlog as of 31/12/2026	Delivery in 2026	Delivery in 2027
Transformer	1. GOV + Local	1213	987	225
Transformer	2. Export	2012	955	1057
Transformer	3. Transformer Service	64	64	
Transformer	Total Transformer	3289	2006	1283
Non-Transformer	4. Steel Structure & Fabricate			
Non-Transformer	5. Distributing Derrick / Arial Truck / O&M / SMA	190	190	
Non-Transformer	Total Non-Transformer	196	196	
Total		3485	2202	1283

Source: TRT

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Risks: The main risk is an easing of US-China geopolitical tensions that allows global buyers to source more transformers from China again. Given China's large manufacturing capacity, a return of Chinese supply could reduce the shortage that is currently pushing orders toward Korea, Japan and Thailand, limiting TRT's export upside.

Price objective basis & risk

Hana Microelectronics (XHNNF)

Our Bt42 PO is based on 31.6x forward P/E (+1SD of historical band) applied to our 2027E EPS, using 2027 as the normalized earnings anchor after IC recovery gains traction. We exclude Phononic contribution, which remains upside optionality. Downside risks are slower IC recovery from geopolitical disruption or inventory overhang, PMS qualification delays, unexpected U.S. tariff changes, and FX volatility. Upside risks are faster IC utilization recovery, earlier PMS breakeven, and Phononic revenue contribution ahead of expectations.

Analyst Certification

I, Suppapong Iemkongaek, CFA, CMT, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

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APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	AAC Technologies	AACAF	2018 HK	Katherine Zhu
	Asia Vital Components	AVMPF	3017 TT	Doris Kao
	BizLink	BIZLF	3665 TT	Doris Kao
	BOE Technology Group Co., Ltd	XCJQF	000725 CH	Brad Lin
	Chenbro	CEOMF	8210 TT	Doris Kao
	Cowell	CWLLF	1415 HK	Katherine Zhu
	Crystal-Optech	XMKAF	002273 CH	Katherine Zhu
	Delta Electronics Inc.	DLTEF	2308 TT	Robert Cheng
	Duksan Neolux	XDSXF	213420 KS	Simon Woo, CFA
	Dynamic Holding	XLHCF	3715 TT	Mike Yang
	E Ink Holdings	PVWIF	8069 TT	Doris Kao
	Elite Material	ETMCF	2383 TT	Mike Yang
	Foxconn Industrial Internet	XDAZF	601138 CH	Robert Cheng
	Goertek	XGKCF	002241 CH	Katherine Zhu
	Gold Circuit	GOCCF	2368 TT	Doris Kao
	Hana Microelectronics	XHNNF	HANA TB	Suppapong Iemkongaek, CFA, CMT
	Hon Hai Precision Industry	HNHAF	2317 TT	Robert Cheng
	Huaqin	XHUAF	603296 CH	Katherine Zhu
	Huaqin	XPEZF	3296 HK	Katherine Zhu
	ISU Petasys	IPTYF	007660 KS	Matt Shin
	Jentech	XVFSF	3653 TT	Doris Kao
	King Slide	KGSLF	2059 TT	Doris Kao
	Kinsus	KNSUF	3189 TT	Mike Yang
	Lens Tech	XLTVF	6613 HK	Katherine Zhu
	LG CNS	XLGLF	064400 KS	Simon Woo, CFA
	LG Electronics	LGEAF	066570 KS	Simon Woo, CFA
	LG Innotek	XLGQF	011070 KS	Simon Woo, CFA
	Lotes	ZYZTF	3533 TT	Doris Kao
	Luxshare	XNJQF	002475 CH	Robert Cheng
	NYPCB	NANYF	8046 TT	Mike Yang
	Quanta Computer	QUCPF	2382 TT	Robert Cheng
	Samsung Electro-Mechanics	SMSGF	009150 KS	Simon Woo, CFA
	Samsung SDI	SSDIF	006400 KS	Simon Woo, CFA
	Samsung SDS	XWPBF	018260 KS	Simon Woo, CFA
	Shennan Circuits	XAMKF	002916 CH	Katherine Zhu
	Taiwan Union Technology Corporation	TWUNF	6274 TT	Mike Yang
	Unimicron	UMCRF	3037 TT	Mike Yang
	Universal Display	OLED	OLED US	Simon Woo, CFA
	Victory Giant	XENEF	300476 CH	Katherine Zhu
	Wistron	WICOF	3231 TT	Doris Kao
	Wiwynn	XWIQF	6669 TT	Robert Cheng
	Xiaomi Corporation	XIACF	1810 HK	Robert Cheng
	Yageo	YGEQF	2327 TT	Robert Cheng
	Zhen Ding Tech	XFAGF	4958 TT	Doris Kao
	Zhongji Innolight	XZICF	300308 CH	Katherine Zhu
	ZTE Corporation	SHZZF	000063 CH	Katherine Zhu
	ZTE Corporation	ZTCOF	763 HK	Katherine Zhu
NEUTRAL				
	Advantech	ADTEF	2395 TT	Robert Cheng
	Asustek	AKCPF	2357 TT	Robert Cheng
	AU Optronics	AUOTF	2409 TT	Brad Lin
	AU Optronics	AUOTY	AUOTY US	Brad Lin
	Auras Technology	AUTCF	3324 TT	Doris Kao
	BYD Electronic	BYDIF	285 HK	Katherine Zhu
	DS Precision	XBASF	002384 CH	Katherine Zhu
	Innolux Corporation	CMLXY	3481 TT	Brad Lin
	Largan Precision	LGANF	3008 TT	Robert Cheng
	Lenovo Group	LNVGF	992 HK	Robert Cheng
	Lenovo Group	LNVGY	LNVGY US	Robert Cheng
	Lite-On Tech	LOTZF	2301 TT	Doris Kao
	Sunny Optical	SNPTF	2382 HK	Katherine Zhu
	TCL Technology Group Corp	XTOOF	000100 CH	Brad Lin

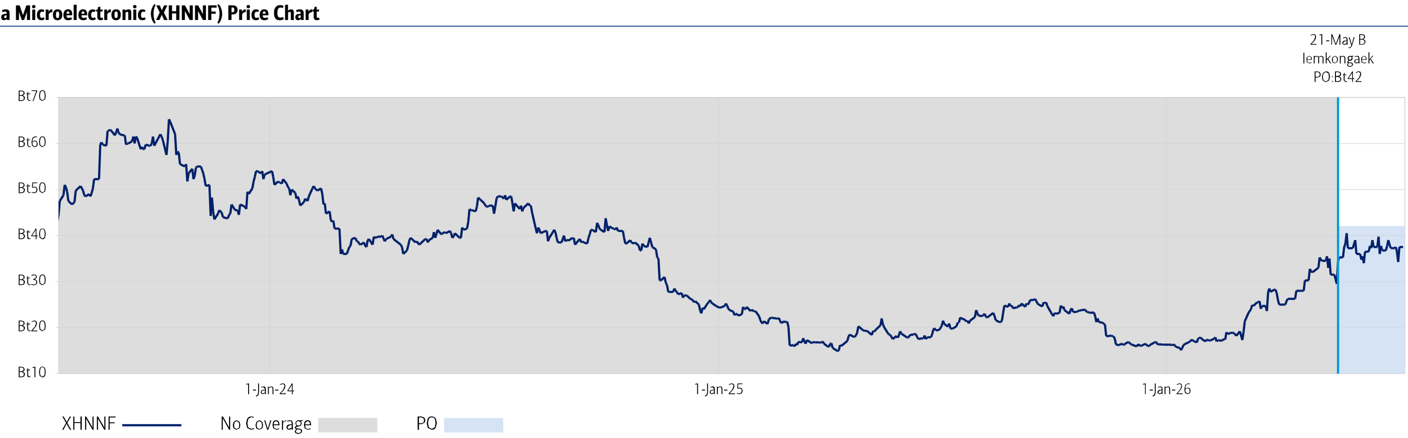
APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
UNDERPERFORM				
	Acer	ASIYF	2353 TT	Robert Cheng
	Catcher Tech	CHERF	2474 TT	Robert Cheng
	Compal Electron	XLCPF	2324 TT	Doris Kao
	Delta Electronics (Thailand)	XLETF	DELTA TB	Suppapong Iemkongaek, CFA, CMT
	Everwin Precision	XSHZF	300115 CH	Katherine Zhu
	Genius Electronic Optical	GNSEF	3406 TT	Robert Cheng
	Lens Tech	XLTGF	300433 CH	Katherine Zhu
	LG Display	LPHLF	034220 KS	Simon Woo, CFA
	LG Display-ADR	LPL	LPL US	Simon Woo, CFA
	Lianchuang Electronic	XWOFF	002036 CH	Katherine Zhu
	Pegatron	PGTRF	4938 TT	Robert Cheng
	Sanan Opto	XEQYF	600703 CH	Katherine Zhu
	Seoul Semiconductor	SLSOF	046890 KS	Simon Woo, CFA
	Shengyi	DGUNF	600183 CH	Mike Yang
	Sunway Communications	XWSVF	300136 CH	Katherine Zhu
	Tianma	XVZLF	000050 CH	Brad Lin
	Transsion	XCINF	688036 CH	Robert Cheng

Disclosures

Important Disclosures

Hana Microelectronic (XHNNF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Electronics Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	34	55.74%	Buy	16	47.06%
Hold	10	16.39%	Hold	3	30.00%
Sell	17	27.87%	Sell	6	35.29%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
Hold	797	22.51%	Hold	496	62.23%
Sell	757	21.38%	Sell	391	51.65%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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